

stages to achieving security in networks

- **Prevention**: stops attacks before they enter system
 - Example: Firewall blocks unauthorized traffic.
- **Detection**: discover the attacks after they have entered
 - Example: IDS detects a suspicious connection and generates an alert.
- **Correction**: involves taking action to address the problem or issue and prevent it from occurring again
 - Example: **Isolate the compromised host**, remove the malware, and fix the vulnerability.
- **Detection** tells you that **something is wrong**; **Prevention** attempts to **stop it**.

security mechanisms

- Security mechanism based on anti-virus, firewall, IDS/IPS
- Anti-virus
- **Viruses** are program which cause **computer failure and damage data**
- In network environment:
 - virus poses an immeasurable threat and can be very destructive
- Anti-virus program are **software** that can be **installed onto** a computer in order to **detect, prevent and make decisions** regarding whether to quarantine or delete malicious programs such as malware, worms or viruses.
- Protect data from modification
- Disadvantage: unable to block **unwanted network traffic** intended to damage the network
- Able to detect the **traffic stable on device**

firewall

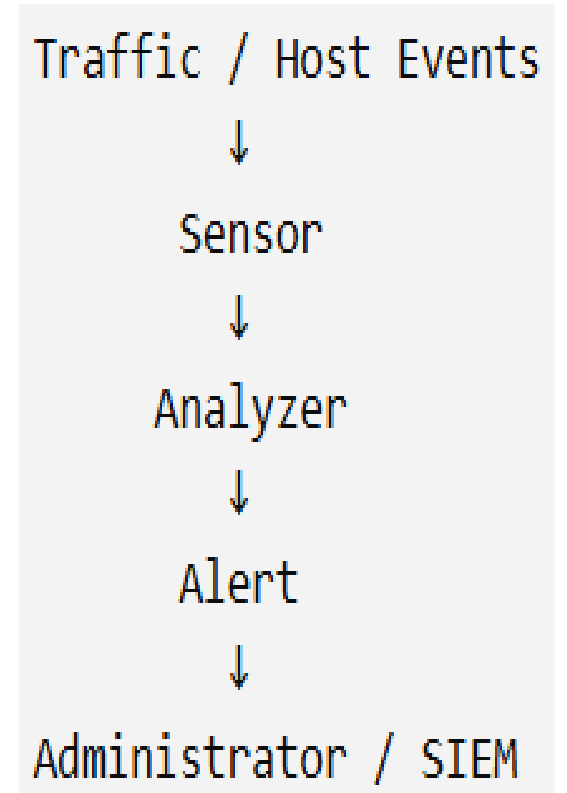
- network traffic is usually **filtered** according to criteria such as **source, destination, protocol or service**
- Firewall prevent from **communication forbidden** by the predefined security policy
- firewall is an active protection
- Firewall **can't detect** the attack which target **legitimate services**
 - firewall to allow access to port 80 (HTTP) on the Internet web server within their network
- Firewall does not examine the **content** of the legitimate traffic

Intrusion detection System(IDS)

- An Intrusion Detection System (IDS) is a network security tool that **monitors network or system activities** for **malicious activities or policy violations**, and produces **alerts** when such **activities or violations** are detected.
- **intruder** is a somebody (hacker or cracker) **attempting to bypass** to system by **authorized or unauthorized** methods in order to **misuse the resources or damaged** the system
- **Intrusions** are **unauthorized access attempts or malicious activities** that are targeted at systems or networks
- **Detection** is the method used to **describe the characteristics** of the analyzer
- **Core Rule:** IDS detects and alerts; it does not normally sit inline to block traffic.

IDS components

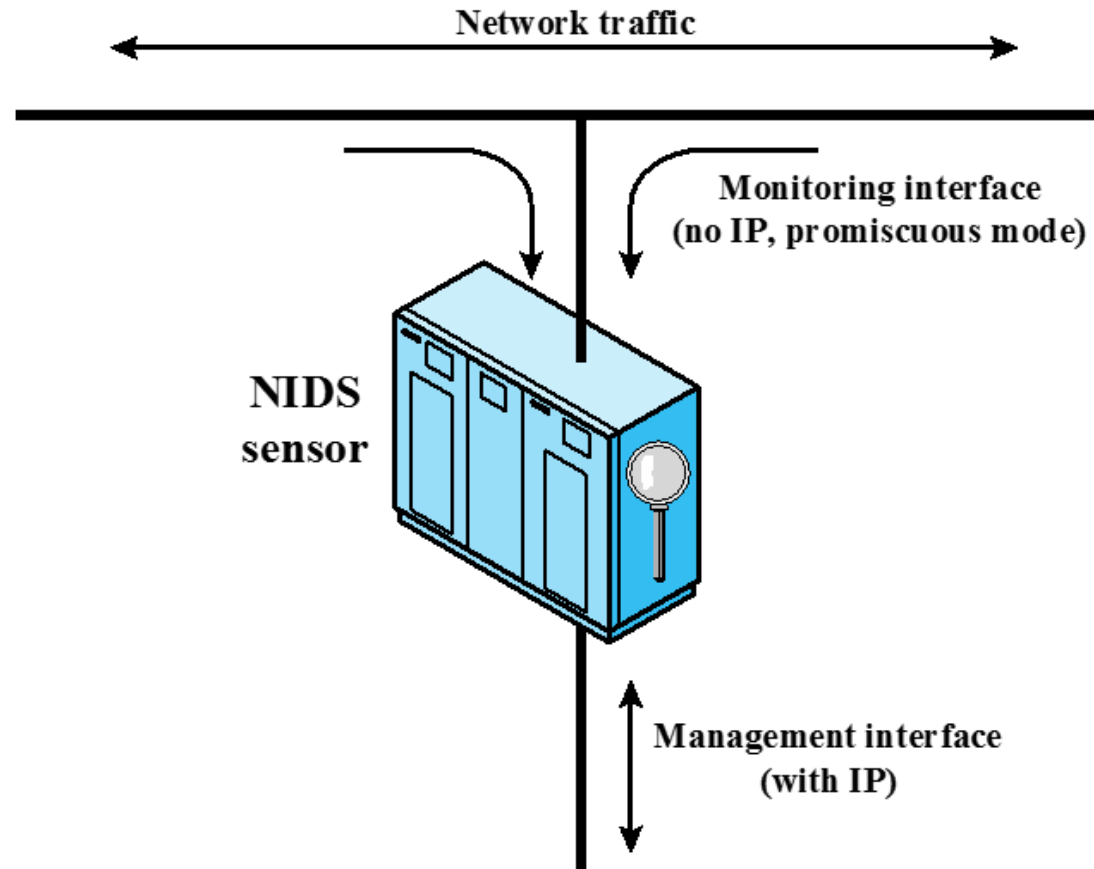
- **Sensors**: collect data from sources
- **Analyzers**: **analyze and decide** if intrusion has occurred
- **User interface**: **view output** and control system behavior



Comparison between IDS And IPS

IPS	IDS
In line	Off line
Detect + block	Detect + alert
Some impact on network (latency, jitter)	No Impact on network (latency, jitter)
Stops pass packets before reach to destination	Response action cannot stop trigger packets, give alert
Sensor issues might affect network traffic	No Network impact if there is a sensor failure
Sensor overloading impacts the network	No network impact if there is sensor overload
Can use stream normalization techniques	More vulnerable to network security evasion techniques

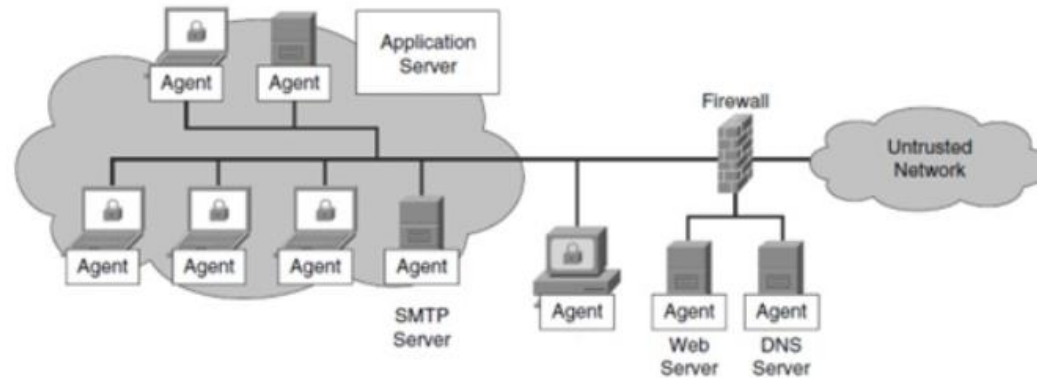
Passive NIDS sensors



Type of intrusion detection

- IDS divided into two basic classes based on **their position**:
 - **host-based IDS (HIDS)** : monitor the **inbound and outbound packet** from that **device only** and **alerts** the user or administrator if **suspicious activity** is detected
 - **Network based IDS (NIDS)**

HIDS

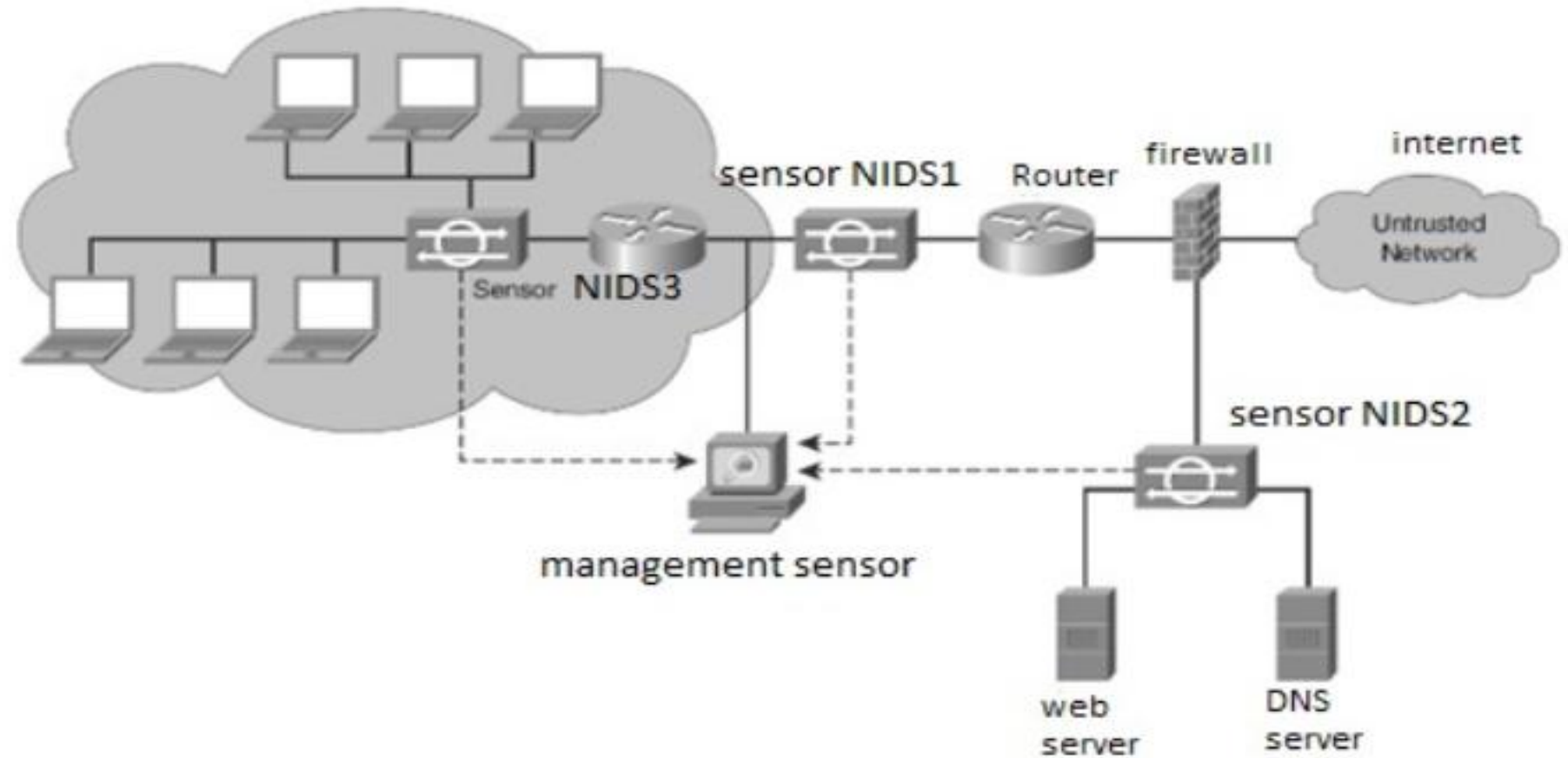


- **HIDS** is a **security system** that **monitors activity** on a **host** and **audit** data such as:
 - resource usage
 - system logs
 - traced by the hosting operating system
- HIDS can **monitor file integrity** by **calculating hashes of critical files** and **comparing** them with a **known-good baseline**.
- **Similar** to the way most **anti-virus** software detects malware

Advantages and disadvantage of HIDSs

- advantage
 - **operate** with **specific host** such as **servers**.
 - **Integrated** with other **network components** and operating system
- Disadvantage:
 - **consume** computer system resources
 - may **conflict** with **existing security policies** (such as firewalls) and operating systems
 - It's **difficult** to analysis intrusion attempts on **multiple computers**
 - **difficult** to maintain in **large networks** with different operating systems and configurations
 - Many **essential** servers **cannot** support this operation

NIDS



- used to analysis traffic at all layers by detecting the behavior of **normal traffic or suspicious activities**

Disadvantage of NIDS

- **unable** to analyses **entire network** packets in **high speed** and high load network
- NIDS itself is **affected** by DoS and DDoS attacks
- NIDS is unable to **inspect** encrypted network traffic

Detection Methods

IDS Detection

- 1. Signature / Misuse-Based Detection
 - Signature Matching / Rules
- 2. Anomaly-Based Detection
 - Statistical
 - Knowledge / Rule-Based
 - Data Mining
 - Machine Learning
- 3. Hybrid Detection
 - Signature + Anomaly

Signature-based SNIDS

- **Signature-based** is a **detection** method that **identifies known malicious** patterns.
- **misuse detection or rule detection**
 - Take **signature** of known attacks and stored it at database
 - Monitor **suspicious packet** and **match** its signature with **database signature**, if matched its hash, or legitimate packet
 - Similar to anti-virus

Advantage and disadvantage of SNIDS

- advantage

- **High confidence** in detection
- Low **false positive** rate
- More understood
- Widely applied

- Disadvantage

- Can not detect **new attack** whose signatures are unknown
- Require a **huge database** that contain, cause overhead in search on signature
- Not appropriate in **high speed** network

Display signature

Home Configuration Event Monitoring Reports Help

CISCO

Configuration > Campus-IPS > Policies > Signature Definitions > sig0 > Active Signatures

SFO-Sensor Corp-IPS Campus-IPS NJ-Sensor Branch-ASA-IPS NY-Sensor Dept-ASA-IPS SJ-ASA-IPS SFO-ASA-IPS LA-Sensor Add Policy Delete Policy Refresh

Signature Definitions

- sig0
 - Active Signatures
 - Adware/Spyware
 - Attack
 - DDoS
 - DoS
 - Email
 - IOS IPS
 - Instant Messaging
 - L2/L3/L4 Protocol
 - Network Services
 - OS
 - Other Services
 - P2P
 - Reconnaissance
 - Releases
 - Viruses/Worms/Trojar
 - Web Server
 - All Signatures

Sensor Setup Interfaces Policies Sensor Management Sensor Monitoring

Edit Actions Enable Disable Restore Default Show Events MySDN Edit Add Delete Clone Export Vid

Filter: Sig ID Filter Clear Signature Wizard

ID	Name	Enabled	Severity	Fidelity Rating	Base RR	Signature Actions	Type	Engine
						Alert and Log Deny Other		
5114/8	WWW IIS Unicode Att...	☒	Medium	85	63	Alert	Default	Service HTTP
5123/2	WWW IIS Internet Pri...	☒	High	85	85	Alert	Default	Service HTTP
5126/0	WWW IIS .ida Indexin...	☒	Medium	100	75	Alert	Default	Service HTTP
5129/0	IOS HTTP Unauth Com...	☒	High	100	100	Alert	Default	Service HTTP
5138/0	Oracle Application Ser...	☒	Medium	85	63	Alert	Default	Service HTTP
5170/0	Null Byte In HTTP Req...	☒	Low	85	42	Alert	Default	Service HTTP
5170/1	Null Byte In HTTP Req...	☒	Low	85	42	Alert	Default	Service HTTP
5177/1	DoS Arnudp	☐	Medium	75	56	Alert	Default	Atomic IP
5183/0	PHP File Inclusion Rem...	☐	Infor...	85	21	Alert	Default	Service HTTP
5188/0	HTTP Tunneling	☒	High	100	100	Alert	Default	Service HTTP
5194/0	Apache Server .ht File...	☒	Low	100	50	Alert	Default	Service HTTP

Total Signatures: 2745 Enabled Signatures: 1161 Active Signatures: 1634 Enabled Active Signatures: 1161

MySDN (Embedded)

Description: Alarms when a URL containing more than 2050 characters is passed to a Oracle server.

Signature ID: 5138/0 Signature Name: Oracle Application Server Shared Library Overflow Severity: Medium

Release Date: 8/11/2010 Release Version: S507

Explanation Related Threats

Cisco.com

Add custom signature

Cisco IPS Manager Express 7.2.3 - Demo Mode

File View Tools Help

Home Configuration Event Monitoring Reports Help

Configuration > Campus-IPS > Policies > Signature Definitions > sig0 > Active Signatures

SFO Sensor Corp IPS Campus IPS N1 Sensor Branch ASA IPS NY Sensor Dept ASA IPS S1 ASA IPS SFO ASA IPS LA Sensor

Add Signature

Name	Value
☐ Signature Definition	
Signature ID	60000
SubSignature ID	0
☒ Alert Severity	Medium
☐ Sig Fidelity Rating	75
☒ Promiscuous Delta	0
☐ Sig Description	
☒ Signature Name	test sig.
☐ Alert Notes	My Sig Info
☐ User Comments	Sig Comment
☐ Alert Traits	0
☐ Release	custom
☐ Engine	Meta
☒ Event Action	Deny Attacker Inline Produce Alert
☐ Swap Attacker Victim	No
☒ Meta Reset Interval	60
Component List	(Click to view or edit the details)
☐ Meta Key	Attacker address
☐ Unique Victims	1
☐ Component List In Order	No
☐ Event Counter	
☐ Event Count	1
☒ Event Count Key	Attacker address

☐ Parameter uses the Default Value. Click the value field to edit the value.
☒ Parameter uses a User-Defined Value. Click the icon to restore the default value.

OK Cancel Help

anomaly NIDS

- known as detection by **behavior**
- **detect deviation** by learn/define **normal behavior**.
- detected by **observing** a **deviation** from **normal or expected behavior**
- If a **deviation is observed**, an **alarm** is generated
- anything that **does not match** to a previously **learned behavior** is considered intrusion

ANIDS

- advantage
 - Can detect previously **unknown behavior** such as **zero-day** attacks
 - It plays a **role** in **flood detection**
- Disadvantage
 - High **false positive** rate
 - Difficulty in identification of **attack type**

Zero-Day Attacks

- A **zero-day attack** is a cyberattack that **tries to exploit software vulnerabilities** that are **unknown or undisclosed** by the software vendor.
- The term **zero-day** describes the **moment** when a **previously unknown threat is identified**

Category virtual sensor 0

The screenshot displays the Cisco Configuration Manager interface for configuring IPS Policies. The breadcrumb navigation shows the path: Configuration > Campus-IPS > Policies > IPS Policies. The left sidebar contains a tree view of configuration categories, with 'Policies' selected. The main content area is divided into two sections: 'Add Virtual Sensor' and 'Event Action Rules "rules0" for virtual sensor "vs0"'. The 'Add Virtual Sensor' section shows a table with one entry, 'vs0', which is assigned to GigabitEthernet0/0.0 (Promiscuous Interface) and GigabitEthernet0/1.20 (Inline VLAN Pair: 20 <-> 40). The 'Event Action Rules' section shows a table with three entries, each with a Name, Enabled status, Sig ID, SubSig ID, Attacker, Victim, Risk Rating, and Actions to Subtract.

Configuration > Campus-IPS > Policies > IPS Policies

SFO-Sensor Corp-IPS Campus-IPS NJ-Sensor Branch-ASA-IPS NY-Sensor Dept-ASA-IPS SJ-ASA-IPS SFO-ASA-IPS LA-Sensor Po < > Add Policy Delete Policy Refresh

IPS Policies

Signature Definitions

- sig0
 - Active Signatures
 - Adware/Spyware
 - Attack
 - DDoS
 - DoS
 - Email
 - IOS IPS
 - Instant Messaging
 - L2/L3/L4 Protocol
 - Network Services
 - OS
 - Other Services
 - P2P
 - Reconnaissance
 - Releases
 - Viruses/Worms/Trojan
 - Web Server

Sensor Setup

Interfaces

Policies

Sensor Management

Sensor Monitoring

Add Virtual Sensor Edit Delete

Video Help

Name	Assigned Interfaces (or Pairs)	Signature Definition Policy	Event Action Override Policy			Anomaly Detection Policy	Description
			Risk Rating	Actions to Add	Enabled		
vs0	GigabitEthernet0/0.0 (Promiscuous Interface) GigabitEthernet0/1.20 (Inline VLAN Pair: 20 <-> 40)	sig0	rules0 (3 action overrides)			ad0	default virtual se...
			HIGHRISK	Deny Packet Inli...	Yes		
				Produce Verbos...	Yes		
			MEDIUMR...	Log Attacker Pa...	Yes		

Event Action Rules "rules0" for virtual sensor "vs0"

Event Action Filters IPv4 Target Value Rating IPv6 Target Value Rating OS Identifications Event Variables Risk Category Threat < > Help

Event Action Filters lets you **subtract** the actions associate with an event if the conditions for that event meet the criteria of the filter.

Add Edit Delete

Name	Enabled	Sig ID	SubSig ID	Attacker (IPv4 / IPv6 / port)	Victim (IPv4 / IPv6 / port)	Risk Rating	Actions to Subtract
Q00000	Yes	5450	0-255	22.214.105.207 0-65535	13.96.99.48 62449	0-100	Produce Alert Produce Verbose Alert
Q00002	Yes	5081	0-255	0.0.0.0-255.255.255.255 0-65535	10.3.3.200 80	0-100	Deny Packet Inline
Q00003	Yes	5450-5460	0-255	22.214.105.207 0-65535	0.0.0.0-255.255.255.255 62449	0-100	Deny Packet Inline Log Attacker Packets

Apply Reset

Virtual Sensor 0

The screenshot shows the Cisco IPS configuration interface for a virtual sensor. The left pane displays the configuration tree with 'Virtual Sensor 0' selected. The main pane shows the configuration details for 'Virtual Sensor 0'.

Configuration > Campus-IPS > Policies > IPS Policies

SFO-Sensor Corp-IPS Campus-IPS NJ-Sensor Branch-A

IPS Policies

- Signature Definitions
 - sig0
 - Active Signatures
 - Adware/Spyware
 - Attack
 - DDoS
 - DoS
 - Email
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 - Instant Messaging
 - L2/L3/L4 Protocol
 - Network Services
 - OS
 - Other Services
 - P2P
 - Reconnaissance
 - Releases
 - Viruses/Worms/Trojan
 - Web Server

Virtual Sensor 0

- Sensor Setup
- Interfaces
- Policies**
- Sensor Management
- Sensor Monitoring

Virtual Sensor 0

Name vs0 **Assigned Interfaces (or Pairs)** GigabitEthernet0/0.0, GigabitEthernet0/1.20

Event Action Rules "rules0"

Event Action Filters lets you:

- Add
- Edit
- Delete

Name	Enabled	Sig ID
Q00000	Yes	5450
Q00002	Yes	5081
Q00003	Yes	5450

Description: default virtual sensor

Interfaces

Assigned	Name	Details
☒	GigabitEthernet0/0	TX (copper)
☒	GigabitEthernet0/1.20	Inline VLAN Pair: 20<->40
☐	GigabitEthernet0/2	TX (copper)
☐	GigabitEthernet0/3	TX (copper)

Select All Assign Remove

Signature Definition

Signature Definition Policy: sig0 ✓

Event Action Rule

Event Action Rules Policy: rules0 ✓

☒ Use Event Action Overrides

Risk Rating	Actions to Add	Enabled
HIGHRISK	☒ Deny Packet Inline (Inline)	☒ Yes
HIGHRISK	☒ Produce Verbose Alert	☒ Yes
MEDIUMRISK	☒ Log Attacker Packets	☒ Yes

Add Edit Delete

Anomaly Detection

Anomaly Detection Policy: ad0 AD Operational Mode: Detect

Advanced Options

OK Cancel Help

IDS / IPS evaluation

- In IDS:
- A **True positive** is an **alert generated** by the **IDS** that is a **legitimate** security incident.
- A **False positive** is an **alert generated by the IDS** that is **not a legitimate** security incident.
- A **False negative** is an **attack** that the **IDS fails** to **detect**.
- A **True Negative** is **legitimate activity** that the IDS **correctly identifies** as **not being a security incident** and **does not generate** an alert.
- Example:
- **False Positive**: Legitimate admin activity → **IDS says "Attack"**.
- **False Negative**: Real attack → **IDS says nothing**.

Detection Trade-off

- **Higher sensitivity** → **More** attacks detected, but **more** False Positives.
- **Lower sensitivity** → **Fewer** False Positives, but **more** attacks may be missed.
- Example:
- **Aggressive detection** → Detect more attacks, but generates more false alarms.
- **Relaxed detection** → Fewer false alarms, but may miss attacks.

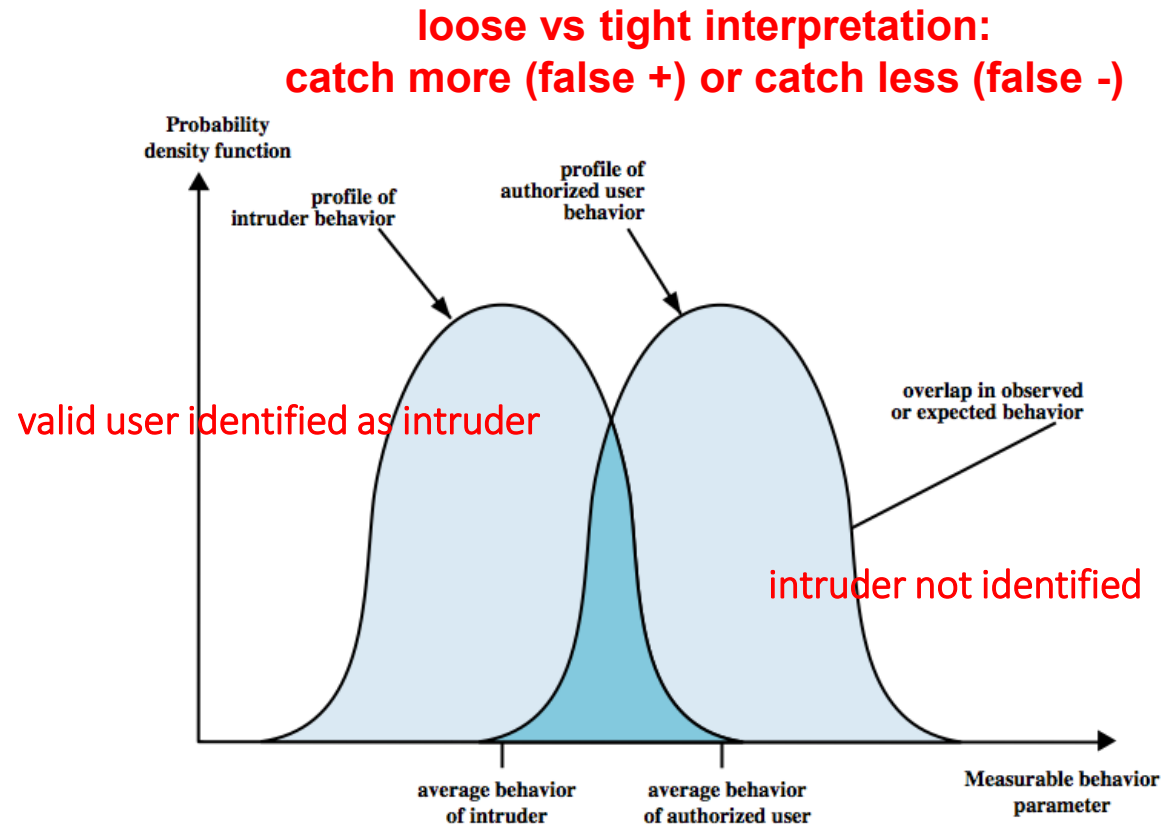
IDS Principles

- Assumption: intruder **behavior** differs from legitimate users

- Expect overlap as shown
- for legit users:

Observe major deviations
from past history

- Problems of:
 - false positives
 - false negatives
 - must compromise



Statistical anomaly-based approach

- A widely used approach for anomaly detection
- user or system **behavior** is **measured** by a **number of variables sampled over time** such as:
 - **Login and logout** time of each session
 - The **resource consumed** duration a **period of time** **processor-memory-disk** during the session
 - Number of **files accessed** in a period of time.

Data Mining-based approach

- Can **support Signature-based Detection** by **discovering** attack **patterns** from large datasets.
- Build **model** from **large data store**, so it's reduced the amount of data that must be reserved
- Advantage
 - **Reduce** the research in large store, **increase detection efficiency**
 - **Reduce** the false alarm
- Disadvantage
 - Analyzed of **large store** database

Knowledge-based detection technique

- Knowledge-based detection uses **predefined rules** to **identify** malicious activity.
- **Rule Sources:**
 - **Expert knowledge** (e.g., signature databases, IDS rules).
 - Observed system behavior patterns.
- **Knowledge Base:** **Built** from information about **known attacks** and **system vulnerabilities**.
- **IDS Role:** IDS store this **knowledge** and **scan for exploitation attempts**.
- **Example:** **Monitoring** for **unusual access patterns** or **unexpected activity**.
- **Hybrid Use:** Often **combined** with **anomaly-based detection** for broader and **more accurate threat detection**.

Knowledge-based detection technique

- Advantage

- Accuracy is good
- Low false alarm rate

- Disadvantage

- requires updated regularly with new vulnerable

Machine Learning-based approach

- Technique which can be capable **knowledge automatically**
- learn from **experience, training**, analytical observation
- The system can continuously **self-improve**
- used in DDoS attack
- Disadvantage:
 - expensive

Traffic analysis

- Packet-Based
- Flow-Based
- Encrypted traffic

packet based NIDS

- Packet-based NIDS: **process** all traffic receive on the **edge router**, and analyze the **whole payload** content **beside** headers
- Packet-based Network Intrusion Detection Systems (NIDS) **analyze individual packets** that are sent across a network in order to detect malicious activity.
- They look for anomalies such as **unexpected data** in a packet, unexpected **protocol** behavior, or unexpected **source and destination** addresses.
- Additionally, NIDSs can be **configured to look** for **specific patterns**, such as **known malicious code signatures**, in order to detect malicious activities.
- **advantage:**
 - all common **kinds of known attacks** and intrusions practically can be detected
- **Disadvantage**
 - cannot detect **unknown attack**
 - Difficulty in work with **high speed network** and vast amount of data

Flow-based NIDS

- **Flow based** is a stream share the following characteristics:
 - IP address, port number source and destination and protocol value
 - provide information about **behavior** of connection
 - attack that only **injected** in **payload** will not be **identified** in flow-based approach
- Flow based can detect attacks such as:
 - DoS, computer worms, network probing , flooding

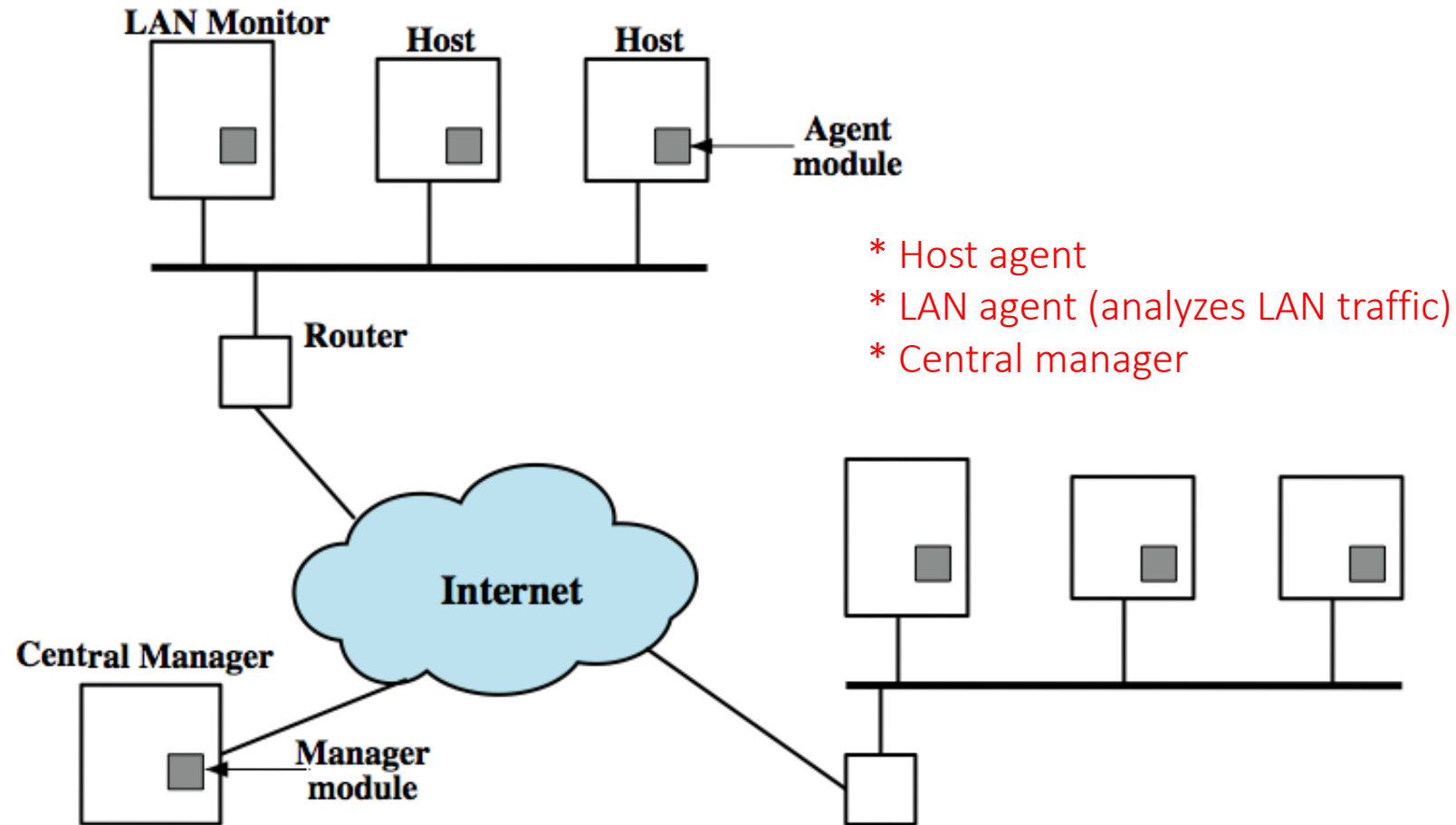
Flow based advantage

- Advantage
- Detect **unknown** attack or zero days
- disadvantage
- Increased **false alarm**

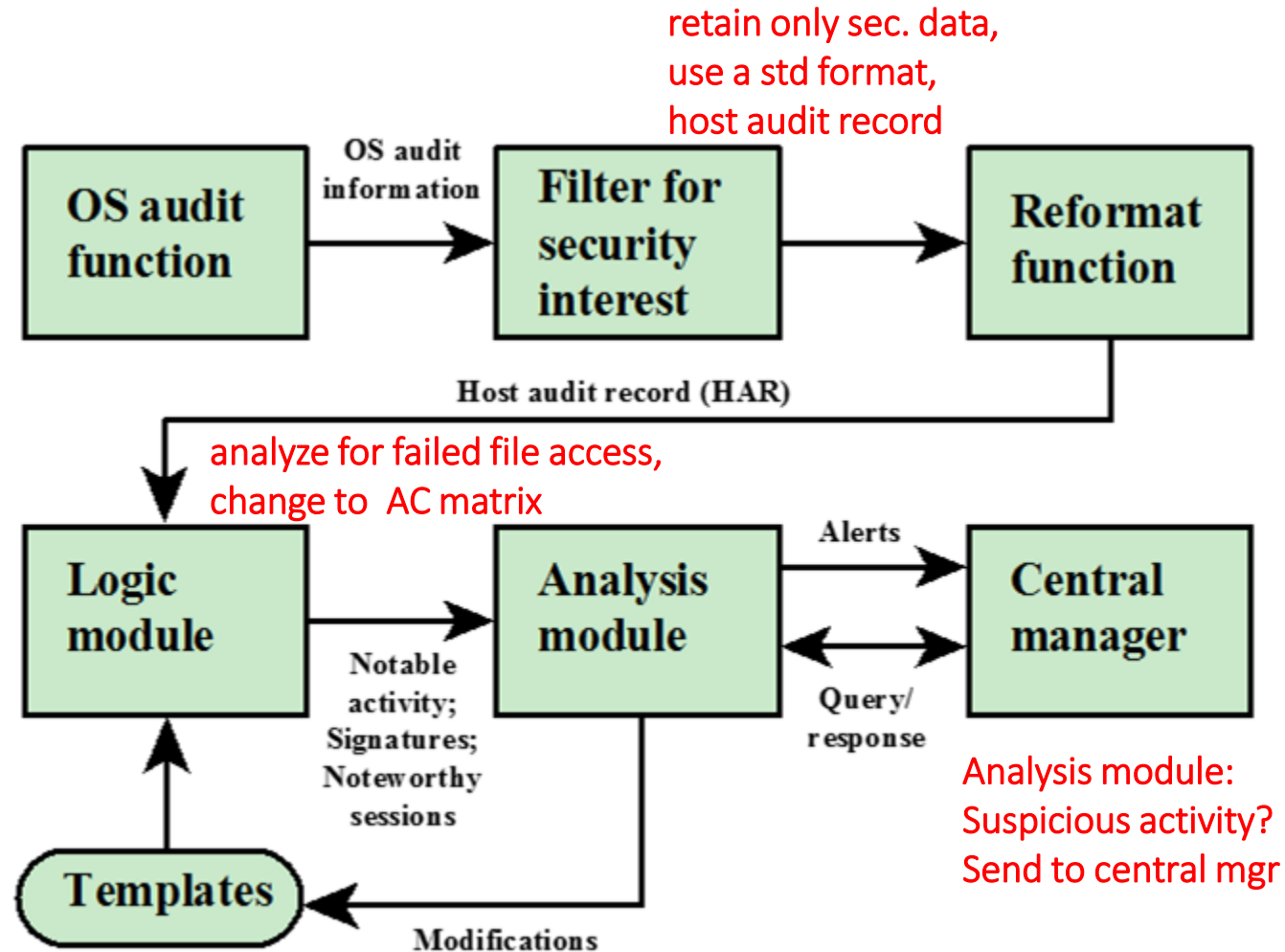
Encrypted Traffic

- NIDS cannot inspect encrypted payload unless traffic is decrypted/inspected at an appropriate point.

Distributed host-based IDS



Distributed host-based IDS: agent architecture



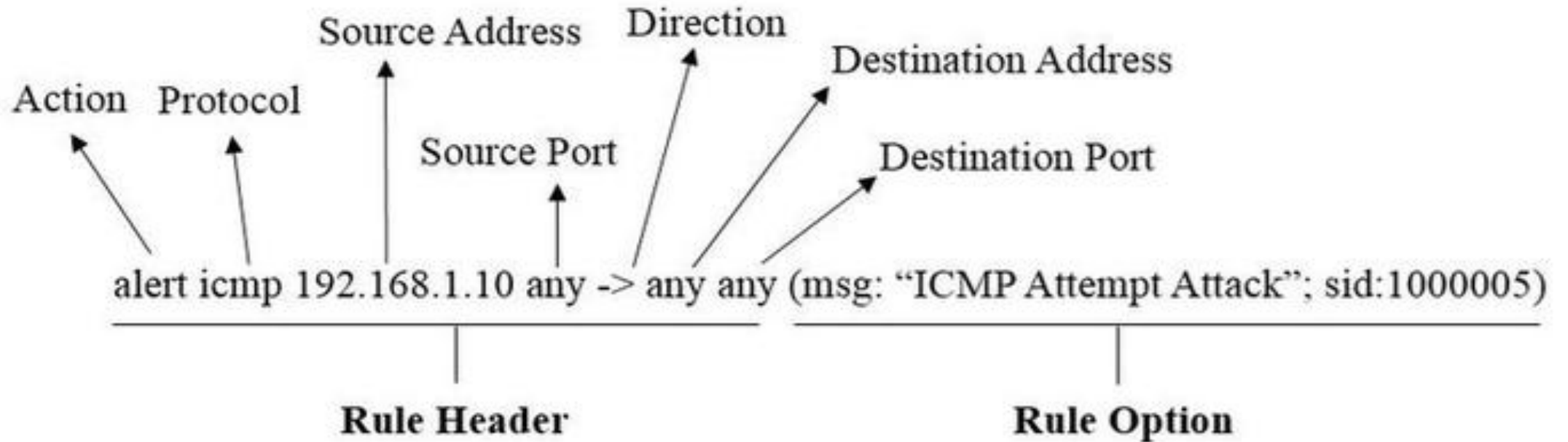
Snort

- Snort is a **famous IDS/IPS** (intrusion prevention system).
- It's **open source**, and you can install it for free in linux.
- **analysis and generates alerts** when **threats are detected on IP networks**
- The **Snort engine runs** in a **virtual service container** on Cisco 4000 Series ISRs
- **Snort IPS Rule Actions**
 - (Snort can be enabled in **IDS** mode or in **IPS** mode).

Snort IDS, IPS

- **Snort IDS mode**
 - **Alert** : Generate an alert
 - **Log** : Log the packet.
 - **Pass** : Ignore the packet.
- **Snort IPS mode**
 - **Drop** : **Block and log** the packet
 - **Reject** : **Block** the packet, **log** it, and then **send** a **TCP reset** if the protocol is TCP or an **ICMP port unreachable message** if the protocol is **UDP**.
 - **Sdrop** : **Block** the packet but **do not log** it

Snort Rule structure



Snort IPS interfaces

- (Snort IPS requires two VPG interfaces)
- **VPG**: virtual port group
- **Management interface (VPG0)**
 - This is the interface that is used to source **logs** to the **log collector** and for **retrieving signature updates** from Cisco.com.
 - For this reason, this interface requires a **routable IP address**.
- **Data interface (VPG1)**
 - This is the interface that is used to **send user traffic** between the Snort **virtual container** service and the **router forwarding plane**.

Practical: Detecting Attacks with Snort

```
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.30.128 netmask 255.255.255.0 broadcast 192.168.30.255
    inet6 fe80::20c:29ff:fe97:2472 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:97:24:72 txqueuelen 1000 (Ethernet)
    RX packets 16817 bytes 24694401 (23.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 7643 bytes 467446 (456.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 248 bytes 82359 (80.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 248 bytes 82359 (80.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions

(kali@kali)-[~]
└─$ sudo leafpad /etc/snort/snort.conf
[sudo] password for kali:
```

Add IP address for kali linux to snort config file

```
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET any 192.168.30.128
```

Snort on kali linux

Open listen state on kali

```
kali@kali:~/log$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i eth0
11/08-15:49:18.785393  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP}
192.168.30.128:40812 → 192.168.30.129:161
11/08-15:49:18.786586  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2]
] {TCP} 192.168.30.128:34902 → 192.168.30.129:705
11/08-15:49:59.722185  [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [
Priority: 3] {UDP} 192.168.30.1:60634 → 239.255.255.250:1900
11/08-15:50:00.722667  [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [
Priority: 3] {UDP} 192.168.30.1:60634 → 239.255.255.250:1900
11/08-15:50:01.723727  [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [
Priority: 3] {UDP} 192.168.30.1:60634 → 239.255.255.250:1900
11/08-15:50:02.724313  [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [
Priority: 3] {UDP} 192.168.30.1:60634 → 239.255.255.250:1900
```

Scan kali from other kali 1

```
root@root:~# nmap 192.168.30.128
```

```
Starting Nmap 6.46 ( http://nmap.org ) at 2022-03-07 14:37 EST
```

```
Nmap scan report for 192.168.30.128
```

```
Host is up (0.0028s latency).
```

```
Not shown: 998 closed ports
```

```
PORT      STATE SERVICE
```

```
23/tcp    open  telnet
```

```
3389/tcp  open  ms-wbt-server
```

```
MAC Address: 00:0C:29:97:24:72 (VMware)
```

```
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds
```

```
root@root:~#
```

SNMP scan on port 161

```
(kali@kali)-[~]  
$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i eth0  
03/07-14:37:31.323713  ** [1:1421:11] SNMP AgentX/tcp request ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:51559 ->  
192.168.30.128:705  
03/07-14:37:31.433128  ** [1:1418:11] SNMP request tcp ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:51559 -> 192.168  
.30.128:161  
█
```

Example 2 scan with scripts

```
root@root:~# nmap -O -A 192.168.30.128

Starting Nmap 6.46 ( http://nmap.org ) at 2022-03-07 14:42 EST
Nmap scan report for 192.168.30.128
Host is up (0.0011s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE      VERSION
23/tcp    open  telnet       Linux telnetd
3389/tcp  open  ms-wbt-server Microsoft Terminal Service
MAC Address: 00:0C:29:97:24:72 (VMware)
No exact OS matches for host (If you know what OS is running on it, see http://nmap.org)
```

Snort activity capture

```
03/07-14:42:31.551331  ** [1:1228:7] SCAN nmap XMAS ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:46292 -> 192.168.30.128:1
03/07-14:42:33.796112  ** [1:1228:7] SCAN nmap XMAS ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:46292 -> 192.168.30.128:1
03/07-14:42:36.037383  ** [1:1228:7] SCAN nmap XMAS ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:46292 -> 192.168.30.128:1
03/07-14:42:39.776121  ** [1:1228:7] SCAN nmap XMAS ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:46292 -> 192.168.30.128:1
03/07-14:42:42.015302  ** [1:1228:7] SCAN nmap XMAS ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.30.141:46292 -> 192.168.30.128:1
```

Example 2: Scan udp port

```
root@root:~# hping3 -2 192.168.30.128
HPING 192.168.30.128 (eth0 192.168.30.128): udp mode set, 28
tes
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1653 seq=0
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1654 seq=1
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1655 seq=2
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1656 seq=3
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1657 seq=4
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1658 seq=5
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1659 seq=6
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1660 seq=7
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
status=0 port=1661 seq=8
ICMP Port Unreachable from ip=192.168.30.128 name=UNKNOWN
```

Snort capture UDP port scan

```
03/07-14:46:42.267053  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1653 -> 192.168.30.128:0
03/07-14:46:43.270386  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1654 -> 192.168.30.128:0
03/07-14:46:44.271426  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1655 -> 192.168.30.128:0
03/07-14:46:45.273331  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1656 -> 192.168.30.128:0
03/07-14:46:46.273679  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1657 -> 192.168.30.128:0
03/07-14:46:47.274995  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1658 -> 192.168.30.128:0
03/07-14:46:48.276824  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1659 -> 192.168.30.128:0
03/07-14:46:49.279048  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1660 -> 192.168.30.128:0
03/07-14:46:50.280747  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1661 -> 192.168.30.128:0
03/07-14:46:51.281301  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1662 -> 192.168.30.128:0
03/07-14:46:52.282297  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1663 -> 192.168.30.128:0
03/07-14:46:53.283133  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1664 -> 192.168.30.128:0
03/07-14:46:54.283754  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1665 -> 192.168.30.128:0
03/07-14:46:55.284456  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1666 -> 192.168.30.128:0
03/07-14:46:56.284992  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1667 -> 192.168.30.128:0
03/07-14:46:57.286015  [**] [1:525:9] BAD-TRAFFIC udp port 0 traffic [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.30.141:1668 -> 192.168.30.128:0
```

Example 3: ping to kali

```
root@root:~# ping -c 10 -s 12000 192.168.30.128
PING 192.168.30.128 (192.168.30.128) 12000(12028) bytes of data.
12008 bytes from 192.168.30.128: icmp_req=1 ttl=64 time=2.42 ms
12008 bytes from 192.168.30.128: icmp_req=2 ttl=64 time=2.03 ms
12008 bytes from 192.168.30.128: icmp_req=3 ttl=64 time=2.01 ms
12008 bytes from 192.168.30.128: icmp_req=4 ttl=64 time=2.11 ms
12008 bytes from 192.168.30.128: icmp_req=5 ttl=64 time=1.91 ms
12008 bytes from 192.168.30.128: icmp_req=6 ttl=64 time=1.82 ms
12008 bytes from 192.168.30.128: icmp_req=7 ttl=64 time=1.86 ms
12008 bytes from 192.168.30.128: icmp_req=8 ttl=64 time=1.71 ms
12008 bytes from 192.168.30.128: icmp_req=9 ttl=64 time=1.91 ms
12008 bytes from 192.168.30.128: icmp_req=10 ttl=64 time=1.90 ms

--- 192.168.30.128 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9026ms
rtt min/avg/max/mdev = 1.715/1.972/2.426/0.185 ms
root@root:~#
```

Capture ping traffic

```
03/07-14:50:41.807096  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:41.807096  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:42.809945  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:42.809945  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:43.812907  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:43.812907  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:44.814433  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:44.814433  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:45.818168  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:45.818168  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:46.820147  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:46.820147  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:47.823552  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:47.823552  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:48.825566  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:48.825566  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:49.830157  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:49.830157  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:50.833316  [**] [1:480:5] ICMP PING speedera [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.30.141 -> 192.168.30.128
03/07-14:50:50.833316  [**] [1:499:4] ICMP Large ICMP Packet [**] [Classification: Potentially Bad Traffic] [Priority: 2] {ICMP} 192.168.30.141 -> 192.168.30.128
```